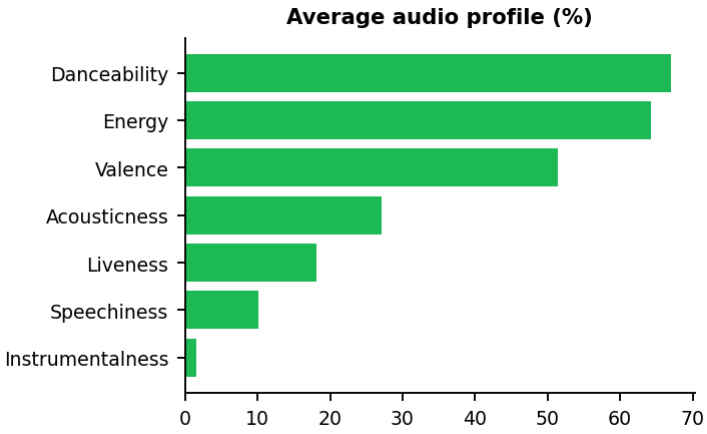
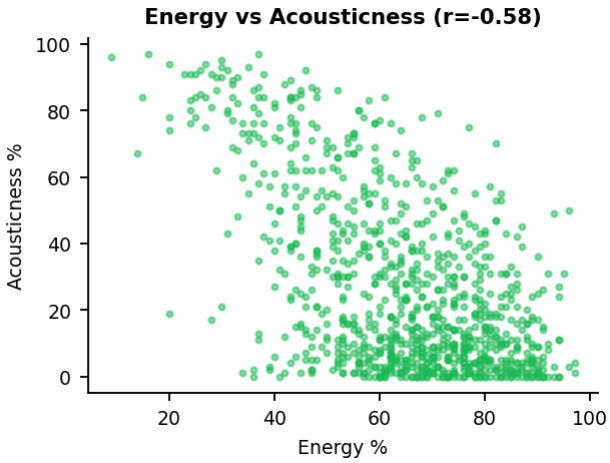
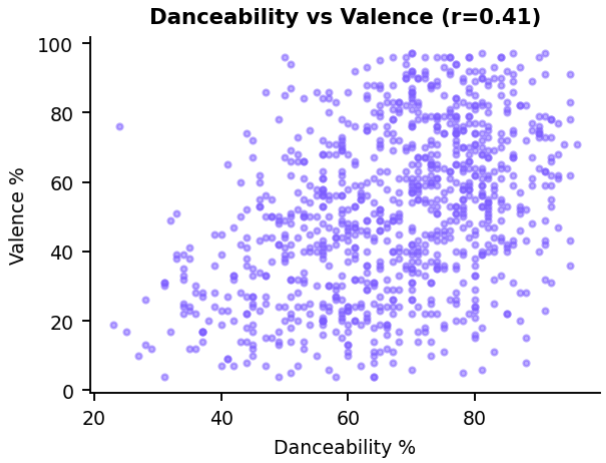
1. Music Analysis

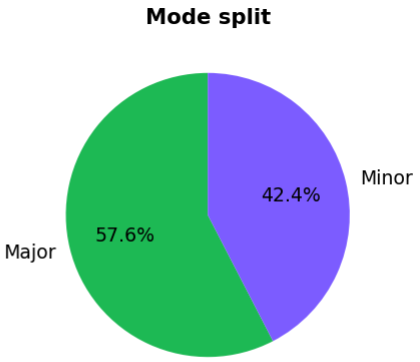
What do popular songs sound like, and which characteristics go together?

948	67%	64%	51%	122
Songs	Avg danceability	Avg energy	Avg valence	Avg BPM









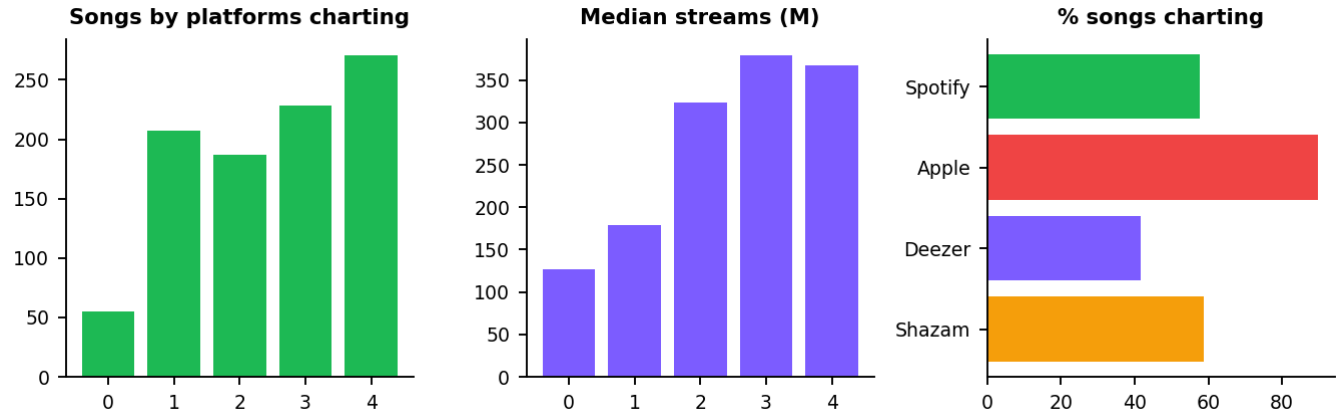
Key insights

- Typical hits are danceable (67%) and energetic (64%) with balanced mood (valence 51%). Instrumentalness averages only 1.6%: hits are overwhelmingly vocal.
- Strongest relationship: energy vs acousticness r = -0.58. Energetic songs are less acoustic.
- Happier songs are more danceable (danceability vs valence r = 0.41) and more energetic (valence vs energy r = 0.36).
- Major mode is 57.6% of songs; minor songs are slightly more danceable (69% vs 65%), a small difference.
- Audio features barely relate to streams (all |r| below 0.12; danceability r = -0.11). Sound alone does not explain popularity. These are associations, not causes.

2. Cross-Platform Presence

Which songs are present everywhere, and how do platforms differ?

271	29%	55	2.48
On all 4 platforms	% on all 4	On no platform	Avg platforms



Top songs on all 4 platforms

Track	Artist	Streams (M)
Sunflower - Spider-Man: Into the S	Post Malone, Swae Lee	2,808
Believer	Imagine Dragons	2,594
Starboy	The Weeknd, Daft Punk	2,566
Seï¿½ï¿½o	Shawn Mendes, Camila Cabello	2,485
lovely - Bonus Track	Billie Eilish, Khalid	2,356
Something Just Like This	The Chainsmokers, Coldplay	2,204

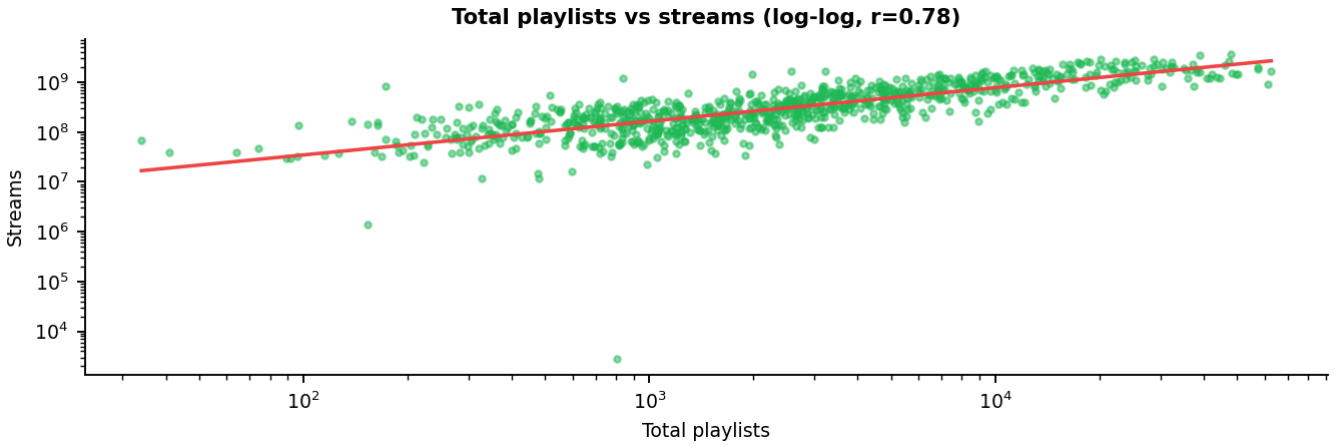
Key insights

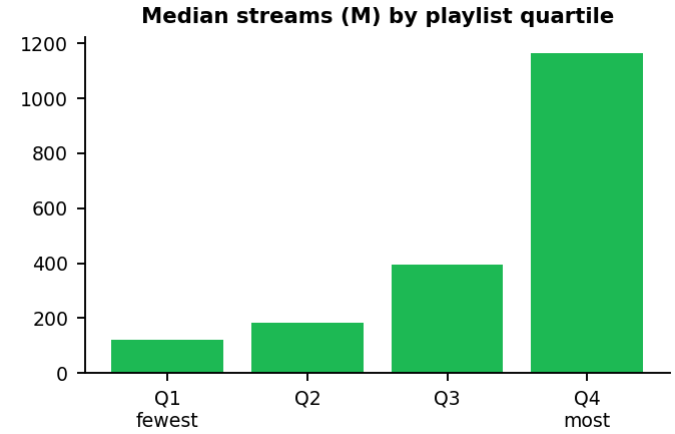
- Apple Music charts 90% of songs; Deezer is the most selective at 42%.
- Songs charting nowhere have a median of 127M streams; songs on 2+ platforms have 323-380M, about 3x higher.
- The benefit flattens after three platforms (median for 4 platforms is not higher than for 3). Association only: charting does not prove it causes streams.

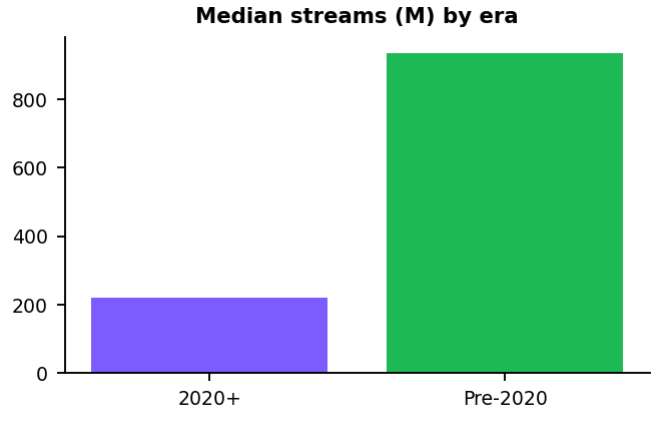
3. Playlist Exposure and Streaming

Do songs in more playlists earn more streams?

0.78	0.83	2,306	288M
Correlation	Rank corr (Spotify)	Median playlists	Median streams







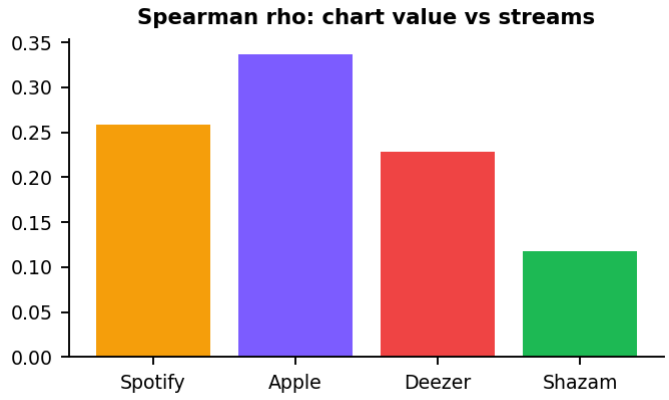
Key insights

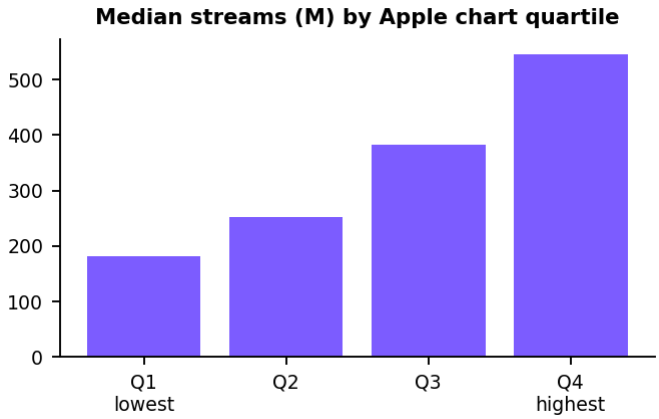
- Strong positive association (r = 0.78); by platform, rank correlation with streams is 0.83 (Spotify), 0.74 (Deezer), 0.67 (Apple).
- Top-quartile songs have a median 1.16B streams vs 123M for the bottom quartile (about 9.4x).
- Pre-2020 songs: median 934M streams; 2020+ songs: 222M. Age is a confounder: older songs had more time to gather playlists and streams.
- Causation cannot be claimed: playlists may boost streams, but popular songs are also added to more playlists (reverse causality).

4. Chart Position and Song Performance

Do stronger chart positions go with higher streams?

0.26	0.34	0.23	0.12
Spotify rho	Apple rho	Deezer rho	Shazam rho





Top songs by streams and chart values

Track	Artist	Streams (M)	Spotify	Apple	Deezer	Shazam
Blinding Lights	The Weeknd	3,704	69	199	20	0
Shape of You	Ed Sheeran	3,563	10	0	7	0
Someone You Loved	Lewis Capaldi	2,887	53	125	0	0
Dance Monkey	Tones and I	2,865	0	167	6	0
Sunflower - Spider-Man: Into the S	Post Malone, Swae Lee	2,808	78	117	4	69
One Dance	Drake, WizKid, Kyla	2,714	24	107	0	26
STAY (with Justin Bieber)	Justin Bieber, The Kid Lar	2,665	36	99	31	0
Believer	Imagine Dragons	2,594	23	121	10	31

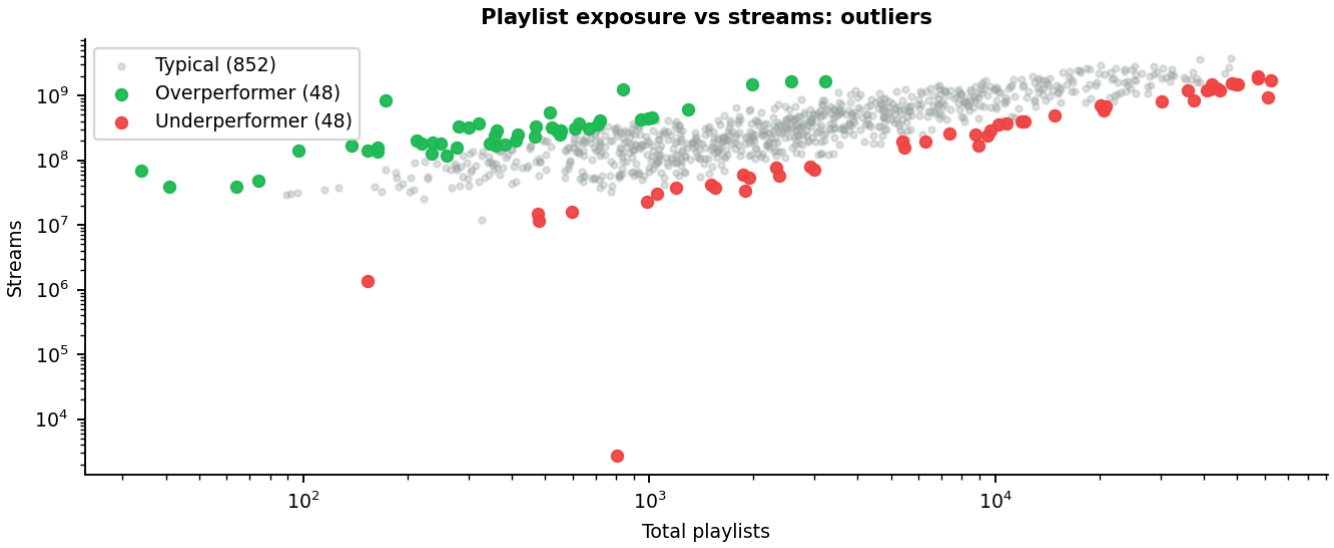
Key insights

- Chart values and streams are positively but weakly related (best: Apple, rho = 0.34); much weaker than playlists (rho about 0.83).
- Apple chart quartile Q4 songs have a median 546M streams vs 182M in Q1.
- Shazam is the weakest signal: it reflects discovery, not repeated listening. Charts show current momentum while streams are lifetime totals, so evergreen hits can lead streams with modest chart values.
- Assumption: chart columns are treated as chart-presence strength (higher = stronger).

5. Outlier and Exceptional Songs

Which songs break the general pattern?

48	48	2022	2012
Overperformers	Underperformers	Median year - over	Median year - under



Top overperformers (high streams, low exposure)

Track	Artist	Year	Streams (M)	Playlists
Dream On	Aerosmith	1973	839	173
Peaches (from The Super Mario Bros	Jack Black	2023	68	34
Enemy - from the series Arcane Lea	Imagine Dragons, Leagu	2021	1,223	840
Apna Bana Le (From "Bhediya")	Arijit Singh, Sachin-J	2022	140	97
Sial	Mahalini	2023	167	138

Top underperformers (high exposure, low streams)

Track	Artist	Year	Streams (M)	Playlists
Que Vuelvas	Carin Leon, Grupo Fron	2022	0	805
Jhoome Jo Pathaan	Arijit Singh, Vishal D	2022	1	153
Get Lucky - Radio Edit	Pharrell Williams, Nil	2013	934	61,316
Mother I Sober (feat. Beth Gibbons	Kendrick Lamar, Beth G	2022	33	1,895
Makeba	Jain	2015	165	8,913

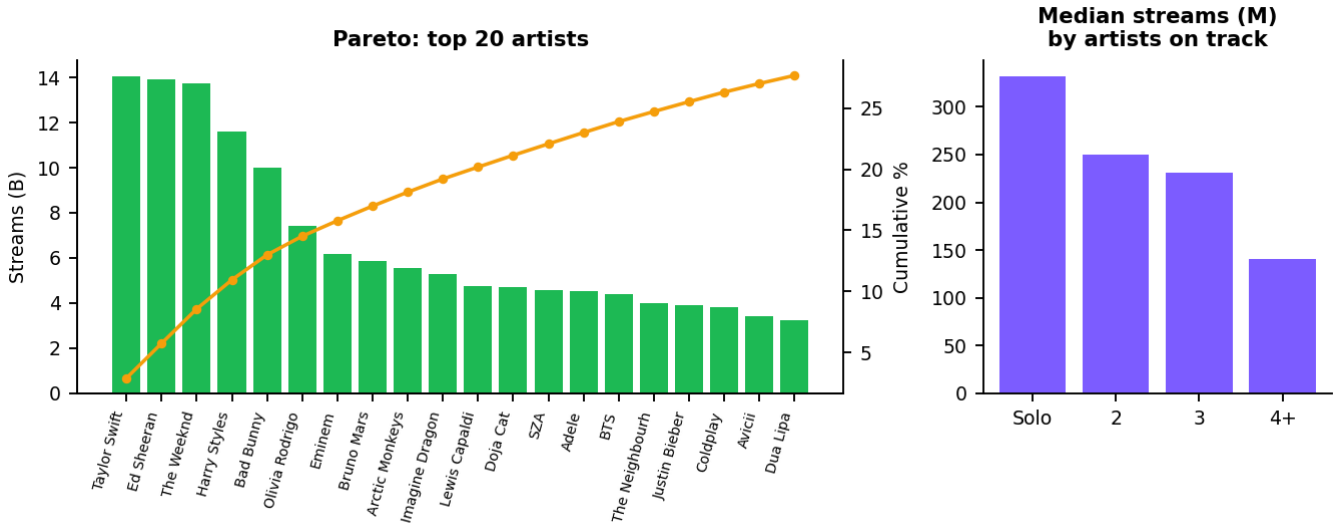
Key insights

- Method: Streams per Playlist = Streams / Total Playlists. Top 5% = overperformers, bottom 5% = underperformers (a simple, explainable but arbitrary cut-off).
- SUPPORTED BY DATA: overperformers skew older or soundtrack tracks; underperformers include seasonal and nostalgic songs with many playlist adds but fewer streams.
- POSSIBLE INTERPRETATION (not proven): overperformers benefit from long-term organic listening, radio, TV/game syncs or algorithmic recommendation; seasonal songs concentrate their streams in a short window.

6. Artist Concentration and 80/20 (Pareto)

Do a few artists account for most streams?

644	217 (34%)	48.6%	66.6%
Artists	Artists for 80%	Top 10% share	Top 20% share



Top 8 artists

#	Artist	Streams (B)	Share %	Cumulative %
1	Taylor Swift	14.05	2.9	2.9
2	Ed Sheeran	13.91	2.9	5.7
3	The Weeknd	13.75	2.8	8.6
4	Harry Styles	11.61	2.4	10.9
5	Bad Bunny	10.00	2.1	13.0
6	Olivia Rodrigo	7.44	1.5	14.5
7	Eminem	6.18	1.3	15.8
8	Bruno Mars	5.85	1.2	17.0

Key insights

- Streams are concentrated but this is not a strict 80/20: the top 20% of artists give 67% of streams and about 34% of artists are needed to reach 80%.
- Top artists: The Weeknd, Taylor Swift, Ed Sheeran, Harry Styles, Bad Bunny. Solo tracks have higher median streams than large collaborations.
- Industry implications: reliance on a few superstars, harder discovery for new artists, and high value of catalog artists. Caveat: the dataset holds only already-popular songs (selection bias), so concentration in the wider industry is likely even stronger. Each credited artist string is treated as one entity.

Methodology and Assumptions

- Data: spotify-2023.csv (953 rows). Cleaning: 1 corrupted streams row removed, 4 duplicate track+artist rows removed (948 songs remain); commas removed from Deezer playlist and Shazam chart values and converted to numbers; missing Shazam values set to 0 (not charting); missing key set to Unknown.
- Calculated fields: Total Playlists (Spotify + Apple + Deezer), Platforms Charting (count of platforms with chart value above 0), Streams per Playlist, Era (Pre-2020 / 2020+), Playlist Band, Mood (valence x energy), Outlier Type (top/bottom 5% of Streams per Playlist).
- Techniques and visual choices: scatter charts for relationships and outliers, column/bar charts for category comparisons, medians instead of means because streams are highly skewed, Pearson correlation for linear association, Spearman rank correlation for skewed and zero-inflated chart data, Pareto chart (columns plus cumulative line) for concentration, KPI cards for headline figures.
- Assumptions: streams are cumulative lifetime totals, so older songs have had longer to accumulate them; chart columns are treated as chart-presence strength; Shazam has no playlist data so it is excluded from Total Playlists; platforms are not weighted by audience size; artist credits are grouped by the full credited string.
- Correlation is not causation. All relationships reported are associations and may be affected by confounders such as song age and existing popularity.
- No forecasting or predictive modelling was used, as instructed. Interactive slicers (era, mode, artist count) are available in the accompanying dashboard.